Portfolio RiskQ

Fast Tail Risk Estimation via Discretized IQAE

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Introduction

Risk teams need reliable VaR/CVaR under tight compute budgets, especially when tail behavior drives capital decisions. We present a classical-to-quantum pipeline that preserves the same loss distribution, tail threshold ℓ, and histogram while enabling IQAE-based tail probability estimation. The goal is a faster tail estimate with clear, auditable outputs that remain interpretable to non-quantum stakeholders.

Background

- VaR = loss quantile at confidence α; CVaR = expected loss beyond VaR.
- Tail threshold ℓ: cutoff defining the tail region used for CVaR and tail probability.
- Portfolio loss: $L = -w^\top r$; tail probability: $p = \mathbb{P}(L \geq \ell)$.
- Discretization: map losses into 2^n bins to create a quantum-ready histogram.
- IQAE: estimates tail probability from the discretized distribution.
- Using the same ℓ and histogram keeps classical vs quantum comparisons fair.

Objectives

- Build a working classical risk engine and dashboard MVP.
- Add discretization and tail-mask generation.
- Deliver a small-scale IQAE simulator with resource reporting.
- Quantify feasibility limits via resource-aware IQAE experiments.
- Communicate results through stakeholder-friendly visuals.

The emphasis is on a complete, inspectable pipeline that supports both classical validation and quantum-oriented estimation.

Models & Materials

Implementation uses Python with NumPy and Pandas for numerical work, a FastAPI backend for serving risk metrics, and a Streamlit + Plotly front end for interactive visualization.

- Gaussian return model; optional crash-mixture scenario.
- Nested Monte Carlo for regime uncertainty.
- Histogram-based discretization for quantum mapping.
- Backend toggle: classical vs quantum comparison path.

Methodology

Nested Monte Carlo (loss samples)

Draw returns from a Gaussian model, then compute portfolio loss as the weighted negative return.

$$r \sim \mathcal{N}(\mu, \Sigma), \quad L = -w^\top r$$

Tail metrics

VaR is the loss quantile at level α; CVaR averages losses beyond VaR.

$$\mathrm{VaR}_\alpha = Q_\alpha(L)$$
$$\mathrm{CVaR}_\alpha = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha]$$

Discretization

- Bin losses into 2^n intervals.
- Produce edges, probabilities, tail mask.
- Bootstrap error estimate for bin stability.
- Bin counts n_i define histogram probabilities p_i; tail probability is the sum over tail bins.

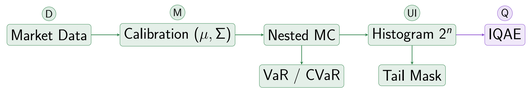
$$p_i = \frac{n_i}{N}, \quad \sum_i p_i = 1, \quad \hat{p} = \sum_{i \in \text{tail}} p_i$$

IQAE (simulator)

- Uses Grover iteration schedule to amplify tail states.
- Returns estimate with confidence interval.
- IQAE probes multiple Grover powers m to infer p from amplified measurements.

$$p = \sin^2(\theta), \quad p_m = \sin^2((2m+1)\theta)$$

System Pipeline (Schematic)



Market data calibrates (μ, Σ), nested MC yields losses, and discretization builds a 2^n histogram and tail mask in the classical pipeline. IQAE then estimates tail probability in the quantum stage.

Analysis & Results

- Validated probability normalization and tail-mask correctness.
- Compared IQAE estimates to true tail probability on toy data.
- Tracked resource usage (qubits, depth, 2Q gates, oracle calls).
- Compared classical vs quantum using the same ℓ and histogram.
- Discretization tests passed (prob sums to 1, mask correctness).
- IQAE toy test estimates matched true tail probability.
- Dashboard showed VaR/CVaR and tail-highlighted histogram.
- Backend toggle enabled classical vs quantum side-by-side comparison.
- Difference fields (abs/rel error) were reported in API/ UI.

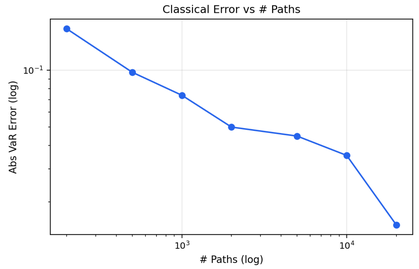
Key Contribution

We present a classical-to-quantum VaR/CVaR workflow that uses a discretized histogram and tail mask to enable IQAE comparison against nested Monte Carlo. We also map feasibility boundaries for NISQ constraints and validate the pipeline in a production-style dashboard.

Experiment Visuals

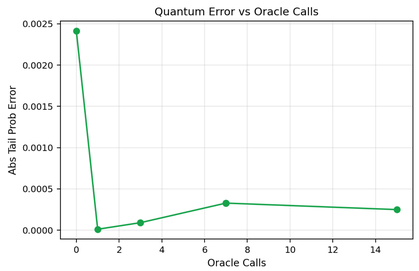
These plots summarize accuracy tradeoffs across classical sampling, quantum query cost, discretization resolution, and noise sensitivity. We report absolute tail-probability error against a known baseline, highlighting where additional compute yields diminishing returns and where noise dominates estimation error.

Classical Error vs Paths



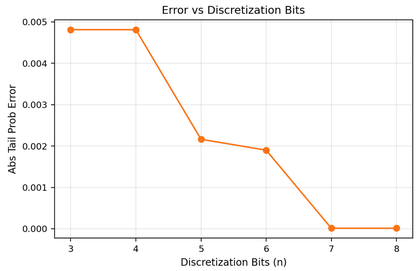
Takeaway: *VaR error falls as $1/\sqrt{N}$; 10k+ paths stabilizes tails.*

Quantum Error vs Oracle Calls



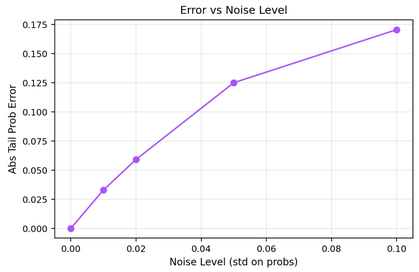
Takeaway: *Sub-1% tail error in a few Grover rounds.*

Error vs Discretization Bits



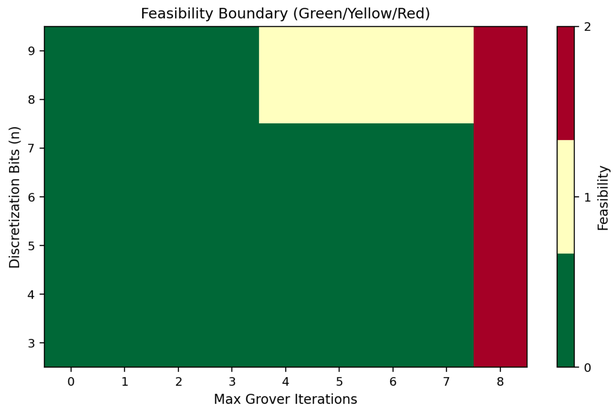
Takeaway: *Gains taper beyond 6–7 bits (binning bias).*

Error vs Noise Level



Takeaway: *Noise ($\sigma \gtrsim 0.05$) dominates error.*

Feasibility Boundary

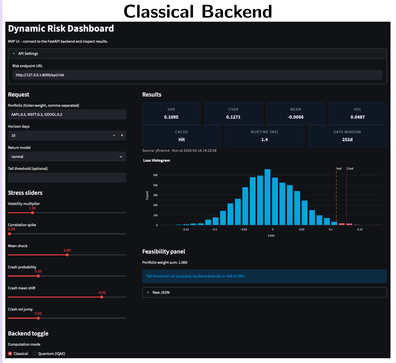


In the feasible region, resource demands stay low enough for near-term devices, while the borderline region signals settings that quickly become impractical. The not-viable region reflects NISQ-era limits where IQAE cost dominates the workflow. Bands use a composite resource score (qubits + depth + oracle calls).

Dashboard Visuals

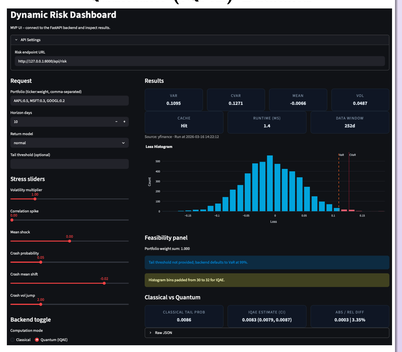
Left panel collects portfolio inputs and stress settings; right panel reports VaR/CVaR, a tail-highlighted histogram, and a classical vs quantum comparison. IQAE reports the same tail probability with a confidence interval and absolute/relative error.

Classical Backend



Interpretability: tail-highlighted histogram with VaR/CVaR markers.

Quantum (IQAE) Backend



Usability: IQAE CI + absolute/relative error vs classical.

Discussion & Implications

- Discretization provides a stable bridge to IQAE.
- Simulator estimates align with expected tail probabilities.
- Resource reports clarify near-term feasibility limits.
- Comparison uses the same threshold ℓ, scenario, and histogram.

Conclusion

The MVP validates a classical-to-quantum risk pipeline with consistent inputs, interpretable outputs, and a clear audit trail from data to tail estimate.

Acknowledgements & Contact

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https://github.com/amirorfali/Dynamic-Risk-Dashboard